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VALIDATION REPORT

R&D MICROBIOLOGY - CHENNAI

Report Title: Summary Report for Validation and Transfer of Microbial Enumeration Tests of Sugammadex Sodium to Rocky Mount, NC

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1.0 PURPOSE

The purpose of this report is to summarise the results of Microbial Enumeration Tests, encompassing Total Aerobic Microbial Count and Total Combined Yeasts and Molds Count, for Sugammadex Sodium manufactured by TEVA, conducted in accordance with the

2.0 SUMMARY

Sugammadex Sodium is a modified gamma-cyclodextrin compound used as a selective relaxant binding agent for the reversal of neuromuscular blockade induced by rocuronium or vecuronium bromide in adult patients. The drug substance is manufactured by TEVA and is intended for use in the preparation of a sterile injectable drug product.

Microbial Enumeration Tests (MET) for Total Aerobic Microbial Count (TAMC) and Total Yeast and Mold Count (TYMC) are required for Sugammadex Sodium in accordance with Pfizer internal policy standard V7112. As no compendial monograph (USP/EP/BP/JP) mandates a specific microbial limit test for this drug substance, the inclusion of TAMC and TYMC testing is driven by Pfizer policy. The validation was conducted to confirm the suitability of the rinse scheme and membrane filtration method for routine microbial enumeration testing.

The validation was performed at the Chennai Microbiology R&D originating laboratory using three batches of Sugammadex Sodium, in accordance with USP <61> and USP <1227>. The validated method was subsequently transferred to the Rocky Mount, NC QC Microbiology laboratory.

3.0 MICROBIAL ENUMERATION TEST METHOD TRANSFER

The Microbial Enumeration Test (MET) validation for Sugammadex Sodium API, manufactured by TEVA, was performed at Chennai R&D Microbiology on three lots of drug substance using the membrane filtration technique with a 0.45 µm mixed cellulose ester membrane. The receiving laboratory for this method transfer is the Rocky Mount, NC QC Microbiology laboratory.

Sample preparation was performed at a 1:10 ratio, comprising 10 g of Sugammadex Sodium drug substance dissolved in 85 mL of sterile phosphate buffer solution pH 7.2, with the addition of 5 mL of 0.1N HCl. Pre-wetting of the membrane filter was performed with an approximate volume of 25 mL Diluting Fluid A (DFA), followed by a rinse scheme of 3×100 mL aliquots of DFA to adequately flush drug substance residues from the filters.

Per SOP-LAB-03312 section 4.2.2, a formal method transfer study is not required for this method because the test method is very similar to the standard test method

4.0 MATERIALS

Media and Diluents:

Consumables:

Equipment's used:

Microorganisms Used:

The following microorganisms were used as challenge organisms for the validation study.

Bacillus subtilis NCTC 10400 (ATCC 6633)

Pseudomonas aeruginosa NCTC 12924 (ATCC 9027)

Staphylococcus aureus NCTC 10788 (ATCC 6538)

Candida albicans NCPF 3179 (ATCC 10231)

Aspergillus brasiliensis NCPF 2275 (ATCC 16404)

Media and Diluents

Table 4.0.1

Name of media and Diluents	Make	Lot No.	Expiry Date / Use before date
Phosphate Buffer	In House	00715609-0062A	
0.1N HCL	Merck	HC01429260	31-Jan-2023
SCDA Plates	Hi media	MPB0656	31-Jan-2023
SDA Plates	Hi media	MPB0903	31-Mar-2023
Rehydration Fluid	Biomerieux	JDP-057	17-Apr-2023
peptone solution pH 7.0	Inhouse	000417F	
Diluting Fluid A	Inhouse	000417A	
Rehydration fluid	Biomerieux	JCK-057	13-Mar-2023
Soybean Casien Digest Agar Plates	Hi media	MPD0511	28-Feb-2021
Sabouraud Dextose Agar plates	Hi media	MPD0668	31-Mar-2021
Diluting Fluid A	In house	000417B	
Diluting Fluid A	In house	000417C	
Diluting Fluid A	In house	000417D	

Consumables

Table 4.0.2

Consumables	Make	Lot No.	Expiry Date / Use before date
Sterile Container	Tarsons	034BVH291020	28-Oct-2025
Stericup-0.22 Micron filtration system	Merck	MP211505G2	30-Apr-2024
Micropipette tips -10 mL	Eppendorf	H175855H	28-Feb-2023
Syringe filter 0.22 micron	Merck	R0JB89897	31-Jul-2023
Sterile 10mL syringe	Dispovan	225102JF2	31-May-2027
Micropipette tips 2-200 Micro litre	AST	21K22S208-302	28-Feb-2025
0.45 m X 47 mm mixed cellulose ester membrane filters	Merck	F1EB16037	31-May-2023
Sterile Container	Tarsons	056BVH191219	18-Dec-2024
0.22µm Stericup Durapore PVDF membrane	Merck	MP192303G2	30-Jun-2022
Sterile micropipette Tips- 10mL	eppendroff	H175855H	28-Feb-2023
0.45µm Filter Cups-mixed cellulose ester membrane filters	Merck	F0EB63356C	31-May-2021

Consumables	Make	Lot No.	Expiry Date / Use before date
Sterile micropipette Tips-200 µL	Tarsons	206JS-090920	08-Sep-2025

Equipment's used

Table 4.0.3

S. No.	Name of Equipment	Equipment No.	Calibration due date	Performance Verification due date
1	Biosafety cabinet	PCE0811	NA	05 NOV 2022
		ADE251	NA	NA
2	Analytical balance	PCE1057	13-Apr-2023	NA
		ADE278	03-Mar-2021	NA
3	Micropipette	PCE0882	16-Feb-2023	NA
		PCE0835	10-Mar-2023	NA
		ADE403	30-Jul-2021	NA
		ADE407	30-Jul-2021	NA

S. No.	Name of Equipment	Equipment No.	Calibration due date	Performance Verification due date
IKKT-RDE0177	07-Mar-2021	NA		
9	XPERT FILTERED BALANCE SYSTEM	ADE332	NA	NA
10	Vortex mixer	IKKT-RDE0102	NA	NA
11	EZ-FIT Manifold	IKKT-RDE0113	NA	NA

S. No.	Name of Equipment	Equipment No.	Calibration due date	Performance Verification due date
4	EZ-Stream Pump	PCE0887	NA	NA
		IKKT-RDE0108	NA	NA
5	pH meter	PCE1059	16-Mar-2023	NA
		IKKT-RDE0167	27-Jun-2021	NA
6	Master flex Digital Pump Drive	PCE0673	NA	NA
7	Incubators	PCE1088	NA	NA
		ADE263	14-Apr-2021	NA
		ADE264	14-Apr-2021	NA
8	Colony counter	PCE0846	11-Mar-2023	NA
		PCE0848	11-Mar-2023	NA

Microorganisms Used

Table 4.0.4

Name of Microorganism	Make	Lot Number	Expiry Date
Bacillus Subtilis ATCC 6633/NCTC10400	Biomerieux	B6365	02-Nov-2022
Pseudomonas aerugi-nosa NCTC 12924 (ATCC 9027)	Biomerieux	B6456	17-Dec-2022
Staphylococcus aureus NCTC 10788 (ATCC 6538)	Biomerieux	B6502	20-Jan-2023
Candida albicans NCPF 3179 (ATCC 10231)	Biomerieux	B6397	19-Nov-2022
Aspergillus brasiliensis NCPF2275 (ATCC 16404)	Biomerieux	B6412	26-Nov-2022

5.0 METHODS/PROCEDURES

A. Microbial Enumeration Test:

Per protocol simultaneously, 3 batches method validation and qualification study were executed in triplicates and the results were summarized in this report.

Sample Quantity: 10g.

The following sample was taken for validation and Qualification. The drug substance batch details are presented in the table below.

Sample preparation for Microbial Enumeration Test:

Samples of the Sugammadex Sodium drug substances were weighed in separate sterile containers as follows. The sample weights recorded for each container across all three batches are presented in the table below.

The Sugammadex Sodium drug substance solution was prepared at a ratio of 10g of Sugammadex Sodium drug substance in 85 mL of sterile Phosphate Buffer pH 7.2, with the addition of 5 mL of sterile 0.1N HCl. Each sample container was mixed by vortexing until the drug substance was completely dissolved. Following complete dissolution, the obtained solution was filtered through a 0.22 µm filtration system. With this ratio, 10mL of test solution contains approximately 1g of the drug substance.

This 1:10 dilution was termed as Solution-S.

NOTE: Sample solution (1:10) used for Product Negative controls were not sterile filtered as to capture inherit bioburden of the raw material.

3 different Batch number of Sugammadex Sodium Solution-S (10mL) was dispensed separately, and pH was checked, the observed results were within the acceptance criteria of 6-8.

Observed pH values as follows:

The pH values recorded for each batch of Sugammadex Sodium are presented in the table below.

Drug Substance Batches

Table 5.0.1

Drug substance	Supplier	Batch/lot number
Sugammadex Sodium	TEVA	7MPS02844051221
Sugammadex Sodium	TEVA	7MPS02844061221
Sugammadex Sodium	TEVA	7MPS02845860319

Observed pH values

Table 5.0.2

S. No.	Batch Number of the sample	pH of the sample
1	7MPS02844051221	7.333
2	7MPS02844061221	7.336
3	7MPS02845860319	7.335
4	Batch No : 0000192271	7.206
5	Batch No. 0000192355	7.206
6	Batch No. 0000192845	7.208

5.1 PRODUCT POSITIVE CONTROL

The Product Positive Control (PPC) was prepared and tested using the membrane filtration technique as described in the following steps.

5.1.1 Pre-wet three filter membranes with a minimum of 25 mL of Diluting Fluid A and filter the contents.

5.1.2 While the vacuum pump is in the off condition, transfer 10 mL of Solution-S to each filter membrane.

5.1.3 Inoculate the microbial suspension (NMT 100 CFU/0.1 mL) of each challenge microorganism (the volume of the suspension of the inoculum should not be more than 1% of the volume of the diluted product) into the 10 mL of Solution-S and filter the solution. A separate test is required for each microorganism listed in the table of challenge organisms.

5.1.4 RINSE THE FILTERS WITH 3 × 100 ML OF DILUTING FLUID A.

5.1.5 Aseptically transfer the filter membrane containing bacterial cultures onto SCDA plates and incubate at 30–35°C for ≤ 3 days. Repeat steps 5.1.1 to 5.1.4 for yeast and mold cultures and aseptically transfer each filter onto SDA plates and incubate at 20–25°C for ≤ 5 days.

5.1.6 Repeat steps 5.1.1 to 5.1.5 for the remaining two batches of Sugammadex Sodium drug substance.

5.2 PRODUCT NEGATIVE CONTROL

The Product Negative Control (PNC) was prepared to confirm the absence of inherent contamination in the test material.

5.2.1 Pre-wet three filter membranes with minimum of 25 mL Diluting Fluid A and filter the contents.

5.2.2 Transfer 10 mL of un-filtered sample solution to each filter membrane and filter the contents. Rinse the filter with 3 × 100 mL quantity of Diluting Fluid A.

5.2.3 Aseptically transfer the membrane onto SCDA plates and incubate at 30–35°C for NLT 5 days.

5.2.4 Repeat the steps 5.2.1 – 5.2.2 and aseptically transfer the membrane onto SDA plates and incubate at 20–25°C for NLT 7 days.

5.2.5 Repeat the steps 5.2.1 – 5.2.4 for the other 2 batches of Sugammadex Sodium drug substance. Record the results in electronic Lab Notebook (eLN) / Lab notebook.

5.3 BUFFER NEGATIVE CONTROL

The Buffer Negative Control was performed to confirm the absence of microbial contamination in the diluent (Phosphate Buffer pH 7.2).

5.3.1 Pre-wet three filter membranes with minimum of 25 mL Diluting Fluid A and filter the contents.

5.3.2 Transfer 10 mL of sterile Phosphate Buffer pH 7.2 to each filter and filter the contents. Rinse the filter with 3 × 100 mL quantity of Diluting Fluid A.

5.3.3 Aseptically transfer the membrane onto SCDA plates and incubate at 30–35°C for NLT 5 days. Repeat steps 5.3.1–5.3.2 and aseptically transfer the membrane onto SDA plates and incubate at 20–25°C for NLT 7 days.

5.4 PEPTONE NEGATIVE CONTROL

The Peptone Negative Control was performed to confirm the absence of contamination in Diluting Fluid A (DFA).

5.4.1 Pre-wet three filter membranes with a minimum of 25 mL Diluting Fluid A and filter the contents.

5.4.2 Transfer 10 mL of Diluting Fluid A to each filter membrane and filter the contents. Rinse the filters with 3 × 100 mL quantity of Diluting Fluid A.

5.4.3 Aseptically transfer the membrane onto SCDA plates and incubate at 30–35°C for NLT 5 days. Repeat steps 5.4.1 through 5.4.2 and aseptically transfer the membrane onto SDA plates and incubate at 20–25°C for NLT 7 days. Examine plates for the presence or absence of microbial growth and record the results in the electronic Lab Notebook (eLN)/Lab notebook.

5.5 INOCULUM COUNT VERIFICATION

Inoculum Count Verification was performed for each challenge microorganism to confirm the inoculum count did not exceed 100 CFU/0.1 mL.

5.5.1 On the day of the study, the inoculum check was carried out for each microorganism by membrane filtration method.

5.5.2 THREE MEMBRANE FILTERS WERE PRE-WETTED WITH A MINIMUM OF 25 ML DILUTING FLUID A.

5.5.3 While the vacuum pump was in the off condition, 100 mL of Diluting Fluid A was added and the microbial suspension (NMT 100 CFU/0.1 mL) of Bacillus subtilis was inoculated and the solution was filtered.

5.5.4 Steps 5.5.2 to 5.5.3 were repeated for the remaining microorganisms. Filter membranes containing bacterial cultures were aseptically transferred onto SCDA plates and incubated at 30–35°C for NLT 3 days. Filter membranes containing yeast and mold cultures were aseptically transferred onto SDA plates and incubated at 20–25°C for NLT 5 days.

5.6 RESULTS

The Microbial Enumeration Test validation was conducted on three batches of Sugammadex Sodium using Soybean Casein Digest Agar (SCDA) plates for bacterial enumeration and Sabouraud Dextrose Agar (SDA) plates for yeast and mold enumeration. Product Positive Control (PPC) and Inoculum Count (IC) plates were evaluated for all five challenge microorganisms across all three batches. The percentage recovery for each microorganism was calculated using the formula: % Recovery = (Average PPC / Average IC) × 100.

5.6.1. Results for Batch number: 7MPS02844051221 (Batch-1)

► Results observed on SCDA and SDA plates:

5.6.2. Results for Batch number: 7MPS02844061221 (Batch-2)

► Results observed on SCDA and SDA plates:

5.6.3. Results for Batch number: 7MPS02845860319 (Batch-3)

► Results observed on SCDA and SDA plates:

From the data presented above, it is observed that the percentage recovery values for all challenge microorganisms across all three batches were within the acceptance criteria of ≥50% and ≤200%, when compared against the respective Inoculum Count (IC) verification plates. The colony morphology observed on PPC plates was found to be consistent with the respective challenge microorganisms.

► Equivalency percentage Recovery Calculation for mold & yeast between SCDA of PPC plate vs SDA of IC Plate:

For Aspergillus brasiliensis and Candida albicans, the PPC plates were prepared on both SCDA and SDA media, and the IC plates were prepared on SDA media. The equivalency recovery comparison between PPC on SCDA and IC on SDA was performed to assess the suitability of using SCDA alone for routine bioburden testing.

Batch Number: 7MPS02844051221 (Batch-1)

Batch Number: 7MPS02844061221 (Batch-2)

Batch Number: 7MPS02845860319 (Batch-3)

The equivalency recovery values for Aspergillus brasiliensis and Candida albicans, comparing PPC on SCDA against IC on SDA, were found to comply with the acceptance criteria of ≥50% and ≤200% for all three batches.

► Results of Product Negative Control on SCDA Plates:

Date of Observation: ⚠ **Not provided in source** .

► Results of Product Negative Control on SDA Plates:

Date of Observation: ⚠ **Not provided in source** .

The Product Negative Control results on both SCDA and SDA plates showed an average count of 0.00 CFU for all three batches, confirming absence of background contamination. The results were found to comply with the acceptance criterion of not more than (NMT) 100 CFU/g.

► Results of Buffer Negative Control and Peptone Negative Control on SCDA Plates:

Date of Observation: ⚠ **Not provided in source** .

► Results of Buffer Negative Control and Peptone Negative Control on SDA Plates:

Date of Observation: ⚠ **Not provided in source** .

The Buffer Negative Control and Peptone Negative Control plates on both SCDA and SDA media showed no growth of microorganisms, confirming the absence of contamination in the diluents and media used during the validation study. All negative control results met the predefined acceptance criteria.

Results observed on SCDA Plates - Batch 7MPS02844051221

Table 5.6.1

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Bacillus sub-tilis NCTC 10400 (ATCC 6633)	44	36	35	38.33	39	35	34	36.00	106
Pseudomonas aerugi-nosa NCTC 12924(ATCC9027)	41	39	42	40.67	40	38	43	40.33	101

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Staphylococc-cusaureus-NCTC 10788(ATCC6538)	46	42	42	43.33	45	43	42	43.67	99
CandidaalbicansNCPF3179 (ATCC 10231)	44	47	42	44.33	46	44	43	44.33	100
Aspergillus brasiliensis NCPF2275 (ATCC 16404)	37	35	36	36.00	36	38	35	36.33	99

Results observed on SDA Plates - Batch 7MPS02844051221

Table 5.6.2

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Candida albicansNCPF3179 (ATCC 10231)				45.00	50	56	54	53.33	84
Aspergillus brasiliensis NCPF2275 (ATCC 16404)				34.00	40	39	36	38.33	89

Equivalency Recovery - Batch 7MPS02844051221

Table 5.6.3

Name of the organisms	Average PPC in SCDA (Average Value)	Average IC in SDA (Average Value)	Percentage recovery
Candidaalbicans	42.67	42	
Aspergillusbrasiliensis	36	33	
CandidaalbicansNCPF3 179 (ATCC10231)	44.33	53.33	83
Aspergillus brasiliensis NCPF2275 (ATCC16404)	36.00	38.33	94

Results observed on SCDA Plates - Batch 7MPS02844061221

Table 5.6.4

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Bacillus sub-tilis NCTC 10400 (ATCC 6633)	41			38.67	39	35	34	36.00	107
Pseudomonas aerugi-nosa NCTC 12924(ATCC9027)	43			41.00	40	38	43	40.33	102

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Staphylococcusaureus-NCTC 10788(ATCC6538)	47			44.00	45	43	42	43.67	101
CandidaalbicansNCPF3179 (ATCC 10231)	43			41.00	46	44	43	44.33	92
Aspergillus brasiliensis NCPF2275 (ATCC 16404)	32			30.33	36	38	35	36.33	83

Results observed on SDA Plates - Batch 7MPS02844061221

Table 5.6.5

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Candida albicansNCPF3179 (ATCC 10231)				44.33	50	56	54	53.33	83
Aspergillus brasiliensis NCPF2275 (ATCC 16404)				32.67	40	39	36	38.33	85

Equivalency Recovery - Batch 7MPS02844061221

Table 5.6.6

Name of the organisms	Average PPC in SCDA (Average Value)	Average IC in SDA (Average Value)	Percentage recovery
CandidaalbicansNCPF3179 (ATCC10231)	41.00	53.33	77
Aspergillus brasiliensis NCPF2275 (ATCC16404)	30.33	38.33	79

Results observed on SCDA Plates - Batch 7MPS02845860319

Table 5.6.7

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Bacillus sub-tilis NCTC 10400 (ATCC 6633)				35.33	39	35	34	36.00	98
Pseudomonas aerugi-nosa NCTC 12924(ATCC9027)				37.67	40	38	43	40.33	93

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Staphylococc-cusaureus-NCTC 10788(ATCC6538)				48.00	45	43	42	43.67	110
CandidaalbicansNCPF3179 (ATCC 10231)				39.67	46	44	43	44.33	89
Aspergillus brasiliensis NCPF2275 (ATCC 16404)				30.33	36	38	35	36.33	83

Results observed on SDA Plates - Batch 7MPS02845860319

Table 5.6.8

Microorgan-isms	Count observed on Product Positive Control (PPC) in CFU				Count observed on Inoculum Count (IC) in CFU				% Recovery (Average PPC/Average IC) x 100
	Plate 1	Plate 2	Plate 3	Average	Plate 1	Plate 2	Plate 3	Average	
Candida albicansNCPF3179 (ATCC 10231)	44	45	46		50	56	54	53.33	86
Aspergillus brasiliensis NCPF2275 (ATCC 16404)	35	34	33		40	39	36	38.33	81

Equivalency Recovery - Batch 7MPS02845860319

Table 5.6.9

Name of the organisms	Average PPC in SCDA (Average Value)	Average IC in SDA (Average Value)	Percentage recovery
CandidaalbicansNCPF3179 (ATCC10231)	39.67	53.33	74
Aspergillus brasiliensis NCPF2275 (ATCC16404)	30.33	38.33	79

Product Negative Control on SCDA Plates

Table 5.6.10

Batch Number of the product	Plate 1	Plate 2	Plate 3	Average
7MPS028440512221	0	0	0	0.00
7MPS028440612221	0	0	0	0.00
7MPS028458603119	0	0	0	0.00
0000192271	0	0	0	0

Product Negative Control on SDA Plates

Table 5.6.11

Batch Number of the product	Plate 1	Plate 2	Plate 3	Average
7MPS02844051221	0	0	0	0.00
7MPS02844061221	0	0	0	0.00
7MPS02845860319	0	0	0	0.00
0000192271	0	0	0	0

6.0 VALIDITY CRITERIA

Based on the 3 batches method validation results, the following validity criteria were found satisfactory as per protocol. The validity criteria and corresponding observed results are presented in the table below.

Validity Criteria

Table 6.0.1

Validity Criteria	Observed Results
The Morphology of the Colonies observed in Product Positive Control (PPC) and Inoculum Count (IC) verification plates were found to be consistent with the challenge microorganisms.	The morphology of the growth observed in the Product Positive Control and Inoculum Count Verification plates were comparable with the challenge microorganism(s).
The Peptone Negative Control and Buffer Negative Control showed no growth of microorganisms.	All negative controls (Buffer Negative Control and Peptone Negative Control) showed no growth.
The Product Negative Control was not more than (NMT) 100 CFU/g. A) The obtained average count for each Batch on SCDA plate showed absence of microorganisms (0CFU/g). B) The obtained average count for each Batch on SDA plate showed absence of microorganisms (0 CFU/g).	No growth was observed in the Product Negative Control on SCDA and SDA plates across all batches tested; the observed Total Aerobic Count met the acceptance criteria of NMT 100 CFU/g.
The Average of Inoculum Count (IC) verification plates was not more than (NMT) 100CFU.	The Inoculum Count verification plate's averages are NMT 100 CFU (highest observed: 53.33 CFU).
The volume of the inoculum did not exceed 1% of the total volume of the diluted product.	The volume of the inoculum has not exceeded 1% of the total volume of the diluted product.

Validity Criteria	Observed Results
The Inoculum Count verification plate’s averages must be NMT 100 CFU.	The Inoculum Count verification plate's averages are NMT 100 CFU (highest observed: 53.33 CFU).
The volume of the inoculum does not exceed 1% of the total volume of diluted product.	The volume of the inoculum has not exceeded 1% of the total volume of the diluted product.

Validity Criteria	Observed Results
The Exposed Settle Plates (Set-1, Set-2 &Set-3) showed No Growth.	Complies with the stated criterion based on the batch results summarized in this report.
The morphology of the growth observed in the Product Positive Control and Inoculum Count Verification plates is consistent with the challenge microorganism(s).	The morphology of the growth observed in the Product Positive Control and Inoculum Count Verification plates were comparable with the challenge microorganism(s).
The Peptone negative control, Buffer negative control must be negative for growth	All negative controls (Buffer Negative Control and Peptone Negative Control) showed no growth.
The Product negative control must be not more than (NMT) 100 CFU/g.	No growth was observed in the Product Negative Control on SCDA and SDA plates across all batches tested; the observed Total Aerobic Count met the acceptance criteria of NMT 100 CFU/g.

7.0 ACCEPTANCE CRITERIA

Microbial Enumeration Test method validation was carried out using 3 batches in triplicate by membrane filtration technique and the outcome is as follows.

The acceptance criteria applied during validation, together with the observed results for each parameter evaluated, are presented in the table below.

Acceptance criteria (Validation)

Table 7.0.1

Acceptance criteria (Validation)	Observed Results
The Recovery of each microorganism in the Inoculated Product Positive Control (PPC) was between 50% and 200%, when compared against the Inoculum Count (IC) verification.	The recovery of each microorganism in the inoculated product (Product Positive Control) was ≥ 50% and ≤ 200% when compared to the inoculum count verification results (observed recovery ranged from 81% to 110% across all batches tested, within the 50–200% range).
For Aspergillus brasiliensis and Candida albicans, Product Positive Control (PPC) and Inoculum Count (IC) was plated both on SCDA & SDA plates. The comparison between PPC on SCDA and IC on SDA for Aspergillus brasiliensis and Candida albicans met the acceptance criteria of 50% and 200%.	Equivalency recovery study between SCDA (TSA) and SDA for growth of yeasts and molds criteria met for Aspergillus brasiliensis and Candida albicans; therefore, the routine bioburden test can be performed using only SCDA (TSA) media at incubation conditions of 30–35°C for not less than (NLT) 3 days.
Hence, the routine bioburden test shall be performed using only SCDA media at Incubation conditions of 30-35°C for NLT 3days.	Complies with the stated criterion based on the batch results summarized in this report.
Acceptance criteria (Validation)	**Observed Results**
If the recovery is <50% or >200%, the results will be reviewed for acceptability and a scientific evaluation should be performed and documented to identify if modifications to the method are required. The test typically must be repeated for the impacted microorganisms to investigate the potential root causes.	The recovery of each microorganism in the inoculated product (Product Positive Control) was ≥ 50% and ≤ 200% when compared to the inoculum count verification results (observed recovery ranged from 81% to 110% across all batches tested, within the 50–200% range).
If the criteria	Complies with the stated criterion based on the batch results summarized in this report.

Acceptance criteria (Validation)	Observed Results
The recovery of each microorganism in the inoculated product (Product positive control) must be □50% and ≤ 200% when compared to the inoculum count verification using the following equationto determine the suitability of the test method: %Recovery = (Average PPC/ Average IC) x 100	The recovery of each microorganism in the inoculated product (Product Positive Control) was ≥ 50% and ≤ 200% when compared to the inoculum count verification results (observed recovery ranged from 81% to 110% across all batches tested, within the 50–200% range).
For Aspergillus brasiliensisand Candida albicans, the Product Positive Controls are plated on SCDA/TSAand SDA:and the Inoculum Checks are also plated onSCDA/TSA andSDA. SDA is the standardgrowth media for mold and yeast. If the comparison between PPC on Version:	Equivalency recovery study between SCDA (TSA) and SDA for growth of yeasts and molds criteria met for Aspergillus brasiliensis and Candida albicans; therefore, the routine bioburden test can be performed using only SCDA (TSA) media at incubation conditions of 30–35°C for not less than (NLT) 3 days.
SCDA/TSA and IC on SDA for Aspergillus brasiliensis and Candida albicans meet acceptance criteria of □50% and ≤ 200%, then the routine bioburden test can be performed using only SCDA/TSA media at incubation conditions of 30-35°C for not less than (NLT) 3 days. This comparison will validate equivalency between SCDA/TSA and SDA for growth of yeasts and molds.	Equivalency recovery study between SCDA (TSA) and SDA for growth of yeasts and molds criteria met for Aspergillus brasiliensis and Candida albicans; therefore, the routine bioburden test can be performed using only SCDA (TSA) media at incubation conditions of 30–35°C for not less than (NLT) 3 days.

8.0 CONCLUSION

Three batches of Sugammadex Sodium were evaluated for Microbial Enumeration Tests (TAMC and TYMC) as part of method validation and transfer to Rocky Mount, NC. The recovery of each microorganism in the Product Positive Control (PPC) was found to be between 50% and 200% when compared against the Inoculum Count (IC) verification for all three batches. All results met the predefined validity and acceptance criteria; on the basis of the data generated, it is concluded that the method is suitable for routine analysis.

For Aspergillus brasiliensis and Candida albicans, the comparison between PPC on SCDA and IC on SDA met the acceptance criteria of ≥50% and ≤200%, thereby demonstrating equivalency between SCDA and SDA for the growth of yeasts and molds. It is therefore concluded that the routine bioburden test may be performed using SCDA media only, at incubation conditions of 30–35°C for not less than (NLT) 3 days.

The following routine test procedure is recommended for Sugamm

9.0 NON-CONFORMANCE

No Non-Conformance was observed during execution of study.

10.0 REFERENCES

The following tables list the references cited in this report.

Table 10.0.1

Document Number	Document Title	Effective/Sign Off Date
USP <61>	Microbiological Examination of Non -Sterile Products	01 MAY 2018
USP <1117>	Microbiological Best Laboratory Practices	01 MAY 2018
USP <1227>	Validation of Microbial Recovery from Pharmacopeial Articles	01 MAY 2018
SOP-LAB-03311	Microbiological Test Requirements for Ingredients Used in the Manufacture of Parenteral Products Procedure, PEH R&D	22 DEC 2020
V7112	Microbiological Methods Validation	12 JUN 2020
L6102	Transfer of test method	08 JUN 2018
SOP-LAB-05213	WRD Biovia Workbook-PCoE R&D	09 OCT2020
SOP-LAB-04874	Validation for Total Aerobic Microbialcount (TAMC) and Total Yeast and Molds count (TYMC) for Active Pharmaceutical ingredients and Excipients –Lake forestand IKKT	04 SEP 2019
SOP-LAB-03312	Biological Laboratory Test Method Validation Procedure, PEH R&D	22 DEC 2020

Document Number	Document Title	Effective/Sign Off Date
SOP-LAB-02515	Generating Scientific Protocols and Reports inGDMS	21 APR2021

Table 10.0.2

Document Number	Document Title	Effective/Sign Off Date
USP <61>	Microbiological Examination of Non -Sterile Products	01 MAY 2018
USP <1117>	Microbiological Best Laboratory Practices	01 MAY 2018
USP <1227>	Validation of Microbial Recovery from Pharmacopeial Articles	01 MAY 2018
SOP-LAB-03311	Microbiological Test Requirements for Ingredients Used in the Manufacture of Parenteral Products Procedure, PEH R&D	22 DEC 2020
V7112	Microbiological Methods Validation	12 JUN 2020
L6102	Transfer of test method	08 JUN 2018
SOP-LAB-02756	Electronic Lab Notebooks-PCoE	29 JUL 2020
SOP-LAB-04874	Validation for Total Aerobic Microbialcount (TAMC) and Total Yeast and Molds count (TYMC) for Active Pharmaceutical ingredi-ents and Excipients –Lake forest and IKKT	04 SEP 2019
SOP-LAB-03312	Biological Laboratory Test Method Validation Procedure, PEH R&D	22 DEC 2020

Document Number	Document Title	Effective/Sign Off Date
SOP-LAB-02515	Generating Scientific Protocols and Reports in GDMS	06 JUL 2020

11.0 CHANGE HISTORY